



# Adaptive correlated Kurtogram and its applications in wheelset-bearing system fault diagnosis

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## ABSTRACT

As one of the crucial subsystems, the performance of the wheelset-bearing system will significantly affect the running safety, stability and comfort of a high-speed train. Compared with the traditional rotating machinery vibration signals, those of wheelset-bearing systems have a distinct difference, for instance, the wheelset tread damage induced vibrations and bearing fault-related vibrations coexist and have similar wave-forms. Therefore, it is necessary to advance a reasonable method to identify the two kinds of *cyclo*-stationary components. To achieve this purpose, the adaptive correlated Kurtogram (ACK) is proposed. The proposed ACK method has two advantages: the first one is it can adaptively generate the paving of the plane for Kurtogram by using scale-space representation (SSR) to guarantee the detected boundaries cover the resonant frequency band well; the second one is it can highlight the special periodic impulses by correlated kurtosis. The scale-scale plane (SSP), which is obtained by SSR, is used to represent the frequency distribution characteristic of the acquired signal, firstly. After that, the paving of the plane is arbitrarily divided based on the SSP. Then the empirical wavelet transform (EWT) is employed to construct the sub-band signals according to the paving of the plane. The correlated kurtosis values with interesting periodic are calculated to distinguish the special repetitive transients. The nodes with the largest correlated kurtosis values are used to conduct a further envelope spectrum analysis. ACK method improves the effectiveness of the identification of the frequency bands containing impact information, and is capable of detecting multiple frequency bands simultaneously. The proposed ACK method is demonstrated by the simulated signal and two experimental signals acquired from a wheelset-bearing system test bench. The results indicate that the ACK can simultaneously extract the wheelset tread damage and bearing fault-related vibration component, and it is superior to the other traditional methods.

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## 1. Introduction

The wheelset-bearing system is one of the core subsystems in a high-speed train. It plays a crucial role in motion transformation and load transmission in the running of a high-speed train [1]. Normally, the main faults of the wheelset-bearing systems include the wheelset-bearing defects and wheelset tread faults [2,3]. If these damages are not detected immediately,

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the safety of the train operations will be threatened [4]. Therefore, it is necessary to monitor the running status of the wheelset-bearing system in real time [4].

Generally, the typical wheelset-bearing system defects consists of cracks, spalls and pitting on the roller, inner race, outer race of bearings as well as the wheelset treads, etc. In pace with the appearance of a defect, either on inner race, outer race or on roller part of bearing or wheelset tread, a series of impacts are generated [5]. With the rotation of bearing and wheelset, repetitive impulse responses are induced by the excitation of the resonances of wheelset-bearing systems, such as the bearing components, axle-box, wheelset and others. The periodic impulses can be collected by vibration sensors installed in the axle-box [6]. To improve the accuracy of fault diagnosis of the wheelset-bearing systems, the periodic impulses should be extracted. Fig. 1 shows a typical wheelset-bearing system, usually, the acceleration sensor is installed on the axle box [7]. However, the weak and periodic impulse response of the defects may be submerged in background noises. Therefore, Wang proposed an effective method to obtain the enhanced envelope spectrum for bearing fault diagnosis [8]. Additionally, the measured acceleration signals usually contain two kinds of cyclic impulse response components: one is bearing fault-related signal, and the other is wheelset tread defect-related signal [9]. Because the two components show almost the same impulsive behavior and have the similar vibration signature, it is a challenge to extract different fault-related impulses from the collected signals. The traditional signal processing methods usually aim to improve the impulsiveness of the signal, but absence of the ability to distinguish two kinds of impulse [10]. As an effective method to realize rotating machine fault diagnosis, envelope spectrum analysis has been widely used [11]. To improve the identification of the fault-related impulse, the collected signals are filtered around the resonant frequency bands. Hence, many advanced methodologies are focused on automatically selecting the reasonable frequency bands. According to the analysis of the collected signals, the resonant frequency bands of the two components are different due to the different vibration transfer paths [12]. Consequently, the faulty parts can be detected by extracting the corresponding resonant frequency bands.

To identify the center frequency and bandwidth of the resonant frequency band, many novel signal processing techniques have been developed [13]. Among them, the spectral kurtosis (SK) has received extensive attention [14]. The SK, as a very powerful tool in replacing the power spectral density, is used to characterize the non-Gaussian component of the signal and locate it in the frequency domain. Subsequently, Antoni presented a comprehensive theoretical framework and a definition of the SK according to normalized fourth-order moment and short-time Fourier transform to reveal the non-stationary signals [15]. Then, many scholars have conducted in-depth research to improve the SK. To realize a multi-fault detection, a method combined with the SK and minimum entropy deconvolution (MED) was proposed [16]. Meanwhile, for effectively suppressing the noise and low-frequency interference, the SK and the PPCA were combined [17]. Although the SK is an effective algorithm to realize bearing fault monitor and diagnosis, it has some drawbacks. For example, the high computation cost has limited its usefulness when a large cycle frequency range is detected. In order to achieve a fast computation, Antoni proposed a fast algorithm based on the finite impulse response (FIR) filter and 1/3-binary tree structure, which is called the fast Kurtogram (FK) [18]. The core of the FK is to establish a two-dimension kurtosis function with respect to the center frequency and bandwidth. The processes of the FK algorithm can be described as two steps: the first step is to obtain the narrow-band signals related to different center frequencies and bandwidths by multiple frequency resolutions; the second step is to utilize the kurtosis indicator to quantify the impulse property of the narrow-band signals. However, two significant shortcomings exist in the FK. Firstly, the fixed frequency resolution of the filtered signal is lack of adaptive property which will impede the accurate detection of the resonant frequency band; Secondly, the kurtosis is sensitive to the outliers which will result in an incorrect frequency band identification. The improvement and optimization of the FK method is addressed in

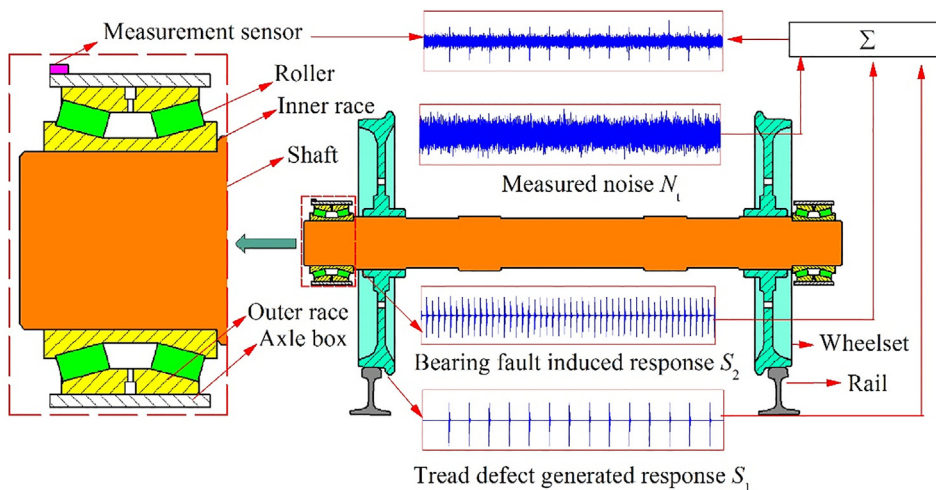


Fig. 1. The wheelset-bearing system.

these two aspects above. Wang [19] utilized a dual-tree complex wavelet transform instead of the FIR filter as well as 1/3-binary structure to decompose the signals. And it can be considered as an effective way to filter out the noise. Lei [20] and Wang [21] applied wavelet packet transform to decompose signals for the generation of the desired narrow-band signals. Discrete quasi-analytic wavelet tight frame (QAWTF) expansion methods are incorporated into a non-dyadic ensemble wavelet subband generating strategy to produce narrow-band signals by Chen [22]. However, the aforementioned four methods still use a fixed frequency resolution which is easy to obtain incorrect frequency band identification results. For realizing the automatic division of frequency bands of Kurtogram, the order statistics filter (OSF) is used to estimate the feature of the Fourier spectrum by [23], and has significantly improved the accuracy of detecting frequency boundaries. Additionally, many developments focus on the indicator to highlight the impulse of the filtered signals. For example, the Infogram [24], the envelope harmonic-to-noise ratio decomposition [25], the Sparsogram [26], the Autogram [27] were proposed to detect the resonant frequency bands of periodic impulses. Although these methods have significant improvement in the heavy interference environment, they also lack the ability to identify a series impulses with special period. As mentioned above, the signals collected from the axle-box contain several repetitive transient components, and that will result in these methods which are unsuitable to realize wheelset-bearing systems diagnosis. Therefore, it is necessary to explore an effective indicator to highlight the special periodic impulse.

Empirical wavelet transform (EWT) is one of the powerful approaches to extract the meaningful modes from the collected signals, and it could be considered as a series of filter banks to decompose a signal into several sub-signals [28]. Scale-space representation (SSR) as a parameter-less method to identify the boundaries of the EWT has attracted lots of attention recently due to its adaptive property [29]. However, the SSR method is prone to obtain an excessively fine frequency partition result for bearing fault diagnosis because an unsuitable scale-space curves (SSCs) selection method is used, and introduced two shortcomings [30]. One is the same modulation information existing in different modes; the other is bearing fault-related information that may be missed in a narrow band mode. To avoid obtaining an unexpected result, some scholars proposed merging the narrow-band signals into a mono-component by ensuring the Pearson's correlation coefficient to the largest [31]. Despite the SSR method is prone to obtain an incorrect detection result, it is still effective to represent the signal frequency characteristic. By selecting a series of effective SSCs, it could obtain a better decomposition result. The SSR method is therefore advised to incorporate the FK method to improve the accuracy of the frequency band division in the proposed method. Meanwhile, a correlated kurtosis indicator is also utilized to indicate the special periodic impulses of the filtered signals [32].

In this paper, a novel adaptive correlated Kurtogram (ACK) is proposed for wheelset-bearing system diagnosis. This method could adaptively generate the paving of the plane based on the SSR which could represent the characteristic of the spectrum of the signal and improve the accuracy of the detected boundaries. The EWT is used to construct the filters based on the obtained paving of the plane to ensure that the lengths of the filtered signals are consistent. Correlated kurtosis indicator is utilized to indicate how much the special periodic fault information is contained in the filtered sub-band signals. As illustrated in the simulation and experimental results, the proposed ACK has a significant improvement in the diagnosis of the wheelset-bearing systems. The paper is arranged as follows: the FK theory and its limitations are discussed in Section 2; the proposed ACK method is established and introduced in Section 3, with the SSR, EWT, and correlated kurtosis being introduced in detail; a complex wheelset-bearing system simulation signal is used to demonstrate the superiority of the proposed ACK method in the detection of the resonant frequency bands in Section 4; and in Section 5, the ACK is applied to deal with the two vibration signals which are acquired from the wheelset-bearing test bench, and the results are compared with the FK and Autogram to illustrate its superiority.

## 2. Fast Kurtogram and its limitations

To realize the wheelset-bearing systems' online diagnosis, the monitor systems should have the characteristics of effectiveness and accuracy, which can simultaneously detect the defects of wheelset-tread and wheelset bearing [6]. Therefore, this article proposed the adaptive correlation Kurtogram (ACK) algorithm, aim to identify the resonant frequency bands of the periodic impulses induced by wheelset-tread defects and wheelset-bearing faults, respectively. Compared with the traditional fast Kurtogram, the proposed method has improved the accuracy of the detected boundaries, and can identify several resonant frequency bands according to the set of the different intervals. In this section, the theoretical background of the FK will be described in detail firstly. Then the three inadequacies of FK method for the diagnosis of the wheelset-bearing systems will be put forward.

### 2.1. A brief introduction to FK

The fast Kurtogram can be considered as a two-dimension kurtosis value function with respect to different center frequencies and bandwidths. The emphasis of realizing Kurtogram fast calculation is to quickly get the narrow-band signals of each level. The FK uses the binary-1/3 tree structure to subdivide the bandwidths into reasonable ratios. The process of the FK can be introduced as follows:

Step 1: Generate the narrow-band signals related to different center frequencies and bandwidths. The original signal is firstly used as the source to be processed, which is named Level 0, and its corresponding frequency band is  $[0, f_s/2]$ , where

$f_s$  is the sampling frequency. The original signal is to be divided into two components: low-frequency and high-frequency, and is named Level 1. The frequency bands of them are  $[0, f_s/4]$  and  $[f_s/4, f_s/2]$ . Meanwhile, Level 0 is divided into three parts in Level 1.6 and the corresponding frequency bands are  $[0, f_s/6]$ ,  $[f_s/6, f_s/3]$ , and  $[f_s/3, f_s/2]$ . After that, the finite impulse response (FIR) filter is employed to generate these five fundamental filters. According to these five filters and the original signal, we could obtain the expected narrow-band signals in Level 1 and Level 1.6. Subsequently, these fundamental filters and original signal are used to obtain the following narrow-band signals in each level. It should be noticed that the down-sampling by factor 2 is conducted for realizing a fast calculation. This manipulation results in the inconsistent lengths of filtered signals in different levels. Finally, we can acquire a frequency division tower as shown in Fig. 2. Obviously, the bandwidth of Level  $k$  is  $(\Delta f)_k = 2^{-k-1}$ .

Step 2: Make the Kurtogram. The narrow-band signal of each node in the paving of the (frequency/frequency resolution) plane is obtained in Step 1. Then the kurtosis values of the complex envelope of the filtered signals are calculated. Concurrently, these values are filled into the corresponding position of Fig. 2 and use various colors to represent their magnitude.

Step 3: Detect the fault related information. Find the node with the maximum kurtosis value for conducting a further envelope spectrum analysis to diagnose the bearing defect.

## 2.2 Shortcomings of the FK algorithm for detecting wheelset-bearing system defects

Despite the undeniable positive aspects of using the FK to identify the resonant frequency band of the fault-related repetitive transients, we find that three vital drawbacks are hidden in wheelset-bearing systems diagnosis:

- (1) The fixed frequency bandwidth and center frequency inevitably appear in the incorrect detection results such as the missing fault signatures and the introduction of much fault irrelevant information as the interference and noise. Fig. 3 compares the ideal boundaries of resonant frequency band and the detected boundaries obtained by the FK [23]. In the Fig. 3, the  $f_s$  is the sampling frequency of the signal. Obviously, the fixed frequency bands cannot cover the resonant frequency band well. This is because of the vibration of mechanical systems are usually highly complex, which makes the resonant frequency locations arbitrary. However, the FK method adopts a binary-1/3 structure to generate the corresponding sub-band signals for realizing a fast computation, therefore, the fixed-resolution approach will cause too many fault-related information being lost or too much irrelevant information entered.
- (2) The kurtosis indicator is only sensitive to the impulsiveness, and is unable to quantify the period of the repetitive impulses. In the FK method, the first step is to obtain a series of sub-band signals, and then the kurtosis values of these sub-band signals are calculated to express the impulsiveness. However, due to the complication of the wheelset-bearing systems, the acquired acceleration signals usually contain several periodic impulses such as the component induced by bearing fault and the component induced by wheelset-tread defect, as shown in Fig. 1. Hence, the kurtosis indicator is improper to identify the different resonant frequency bands of the vibration signals of the wheelset-bearing systems.
- (3) The lengths of filtered signals are inconsistent. Despite the undeniable positive aspects of using binary-1/3 structure and downsampling to achieve a fast computation, it hides a critical shortcoming which is the difference of signal lengths. With the level increases, the length of the filtered signal becomes short, and the FK method may indicate an incorrect frequency band. A white Gaussian noise signal may have a maximum kurtosis value after decomposition [22]. In order to guarantee the effectiveness of the FK, the length of vibration signals should be ensured. However, in many cases, only vibration signals with limited length can be obtained, which may affect the effectiveness of the FK method.

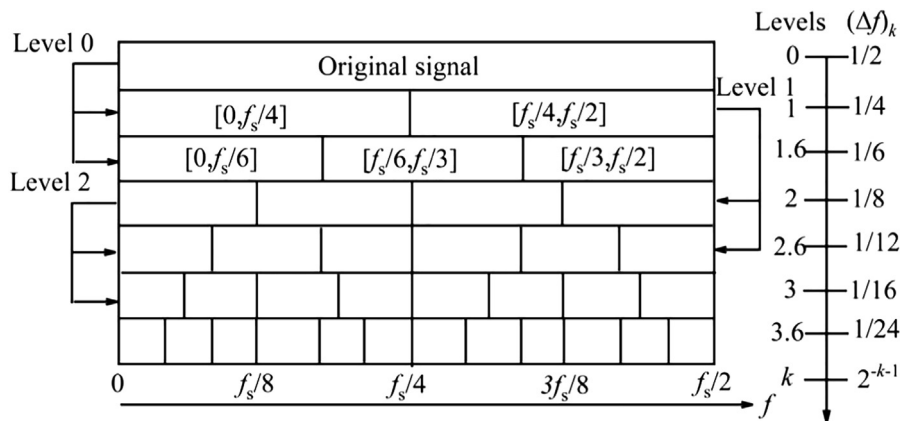


Fig. 2. Frequency-band division using a 1/3-binary tree structure.

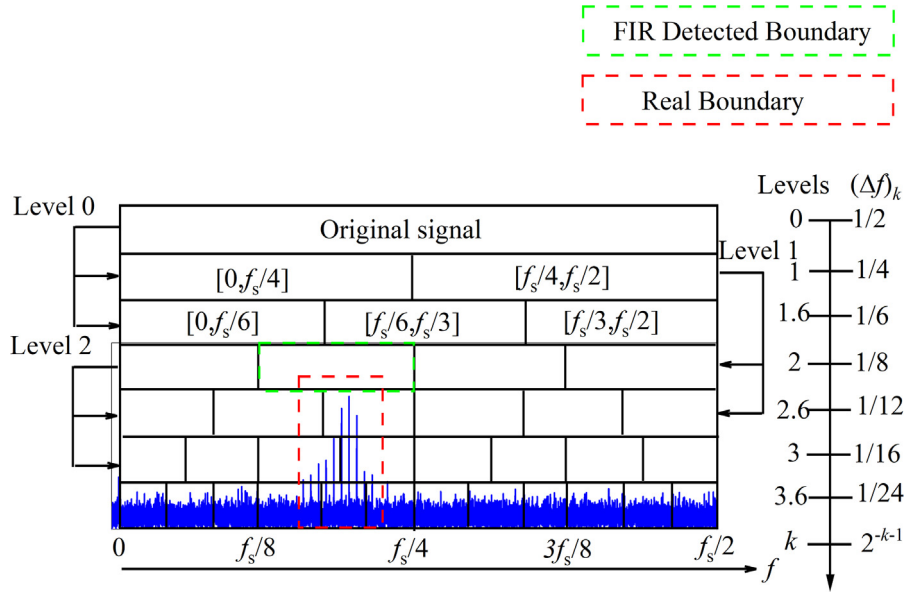


Fig. 3. Spectrum and the detected boundaries by FK.

From the three drawbacks mentioned, we can obtain the following conclusions: The frequency band identified by the FK may not be consistent with the real situation of the resonant frequency band; additionally, the Kurtogram built on the kurtosis indicator cannot detect different resonant frequency bands of the wheelset-bearing systems' vibration signals.

Thus, it is significant to explore a novel algorithm to identify the different resonant frequency bands for achieving wheelset-bearing systems' diagnosis.

### 3. Proposed method

The purpose of this article is to develop a method that can detect the faults of the wheelset-bearing systems through the adaptive correlated Kurtograms. In order to reach this purpose, the FK has been optimized from the aforementioned three drawbacks. The first one is to adaptively generate the paving of the plane according to the signals' frequency domain features. The second one is to filter the signals utilizing the EWT to ensure that the filtered signals have a consistent length. The third one is to indicate the special period impulsiveness of the filtered signals by using the correlated kurtosis indicator. It is noticed that the obtained adaptive correlated Kurtograms are not unique, which benefits from setting different interest impulse intervals. Finally, the frequency bands of bearing fault-related repetitive transients and wheelset-tread defect periodic impulses are separately detected from the adaptive correlated Kurtograms, also the extractions of the fault-related components are achieved. The flowchart of the proposed adaptive correlated Kurtogram has been displayed in Fig. 4, and the details of each step are described as follows.

Step 1: Collect the vibration signal, and generate the paving of the plane

The spectrum  $X(f)$  of the signal  $x(t)$  is calculated through the fast Fourier transform (FFT) algorithm. Then the scale-space plane (SSP) of the spectrum  $X(f)$  is obtained by using the SSR. The frequency domain characteristics of the signal are represented by the SSP. Hence, the frequency boundary division of each level can be adaptively accomplished relating to these characteristics. The SSR-based frequency boundary division method will be explained in 3.1.

Step 2: Decompose the signal with the usage of the EWT

The EWT is an adaptive decomposition algorithm developed by Gilles and can be considered as a series of filter banks [33]. The EWT method uses Meyer wavelet as the basis function to establish the empirical wavelet function and empirical scale function for realizing signal adaptive decomposition. The narrow-band signals of each node in the paving of the plane are obtained by the EWT in this step. And more details of the EWT will be described in 3.2.

Step 3: Calculate the correlated kurtosis, and make Kurtograms

The correlated kurtosis indicator can be used to represent the periodic impulses with a fixed interval. Therefore, it has a strong ability to identify the resonant frequency bands of different faulty parts. The correlated kurtosis values of the filtered signals are filled into the paving of the plane and distinguished by color. More details of correlated kurtosis will be described in 3.3.

Step 4: Find the frequency bands which have the maxima correlated kurtosis values

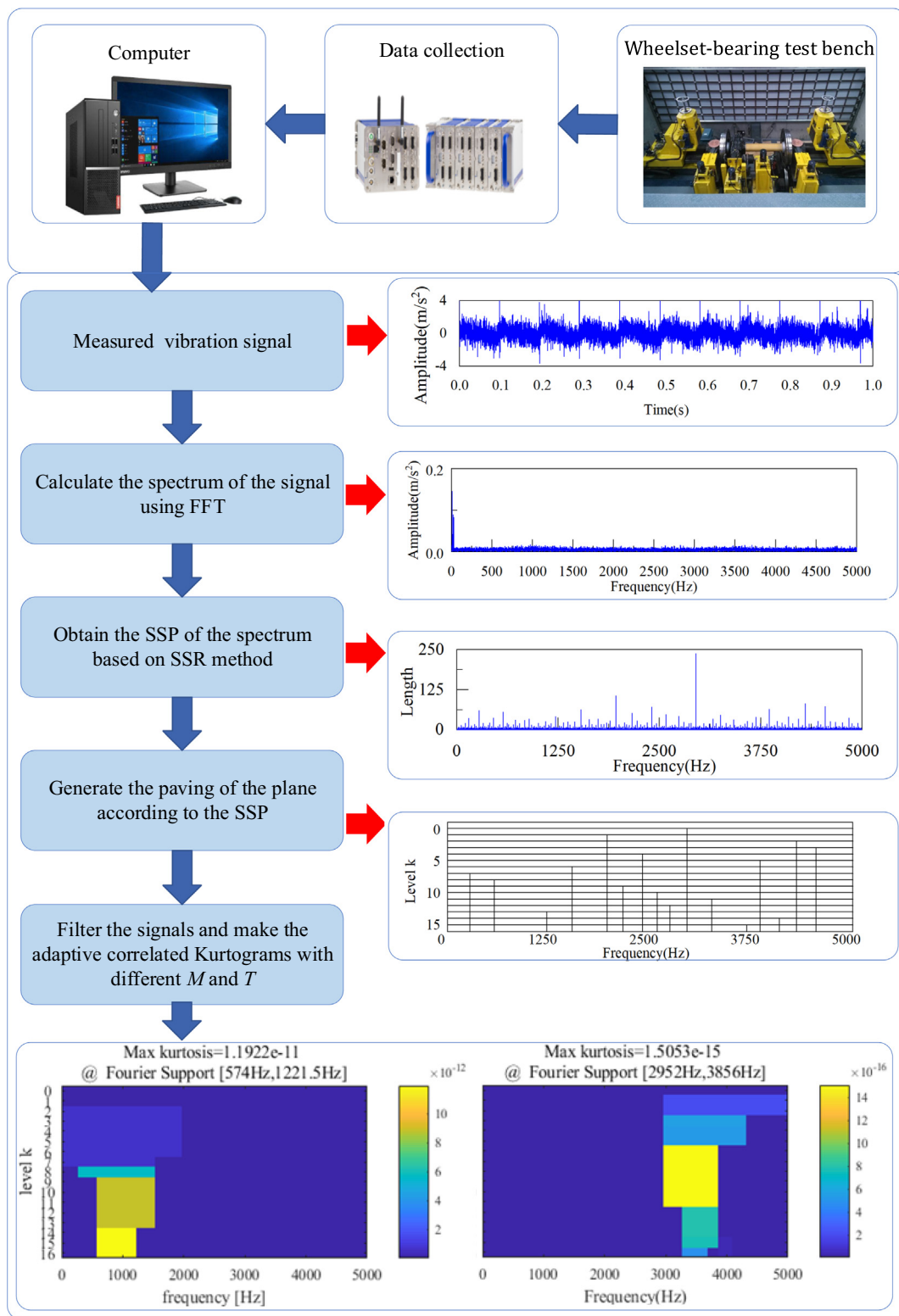


Fig. 4. Flowchart of the proposed ACK method.

The optimal resonant frequency bands of these periodic impulses with different intervals are obtained, and the narrow-band signals are extracted for a further envelope spectrum analysis.

### 3.1.1. SSR-based adaptive the paving of the plane generation method

The FK method is prone to lead an inaccurate identification of the resonant frequency band due to its non-adaptive property. To improve the accuracy of the detection, the generation process of the paving of the plane should be closely related to the signal frequency domain features. To achieve this purpose, the SSR method, which can be interpreted as the global trend of the Fourier spectrum with different scale parameters, is adopted to adaptively generate the paving of the plane. In the proposed method, the SSP needs to be obtained firstly by SSR method. The description of the SSR method is introduced as follows.

$X(f)$  is the Fourier spectrum of the signal  $x(n)$ ,  $L(f, s)$  is defined as the scale-space representation of the  $X(f)$ , and can be expressed as:

$$L(f, s) = X(f) * g(f, s) = \sum_{m=-J}^{+J} X(m-f)g(f, s) \quad (1)$$

where  $g(f, s)$  is the kernel function, and expressed as:

$$g(f, s) = \frac{1}{\sqrt{2\pi}s} e^{-\frac{f^2}{2s}} \quad (2)$$

where  $s$  is the scale parameter;  $J$  denotes the width of the kernel function and is normally considered as  $J = 6\sqrt{s} + 1$  [33].

The number of minima decreases in the scale-space representation  $L(f, s)$  and no new minima appears with the scale parameter or scale step parameter ( $\sqrt{s} = N^2\sqrt{s_0}$ ) \*  $s$  increase. The total number of initial minima in Fourier spectrum is denoted by  $W_0$ . With the increase of the parameter scale, the number of minima of  $L(f, s)$  decreases, and the iteration terminates when the number of minima of  $L(f, s)$  is zero. When the number of minima is zero, it means the  $L(f, s)$  is a convex function which is the same as the kernel function  $g(f, s)$ , and the whole frequency characteristics of the signal can be represented in the SSP. Each minima  $f_i (i \in [1, W_0])$  generates a scale-space curve (SSC)  $C_i (i \in [1, W_0])$  in the scale-space plane (SSP) with length  $R_i (i \in [1, W_0])$  [28]. Hence, the number of SSCs in the SSP is  $W_0$ . After that, we could conceive the  $R_i (i \in [1, W_0])$  as the life span of the  $i$ th minima, and defined it as:

$$R_i = \max\{N/\text{the } i\text{th minimum exists}\} \quad (3)$$

Because the information frequency bands have several highest amplitudes in the spectrum amplitude, the minima around the frequency bands has the larger characteristic length. The corresponding SSCs are longer. According to this feature, the Fourier support of a meaningful mode (i.e. the resonant frequency band) can be obtained by selecting the appropriate SSCs from the SSP [29]. According to that, the SSP is used to adaptively generate the paving of the plane to avoid the incorrect identification results induced by the second shortcoming of the FK mentioned in Section 2.2. The details of this process are introduced as follows.

Firstly, the maximum decomposition level is assumed to be  $K$ . Then SSCs  $C_i (i \in [1, W_0])$  are sorted in a descending order, named  $C_i^d (i \in [1, W_0])$ , according to the length  $R_i (i \in [1, W_0])$ . After that, the longest  $K$  SSCs, named  $C_i^d (i \in [1, K])$ , are selected to adaptively obtain the paving of the plane, and the corresponding frequencies can be denoted as  $f_i (i \in [1, K])$ . Finally, we divided the frequency bands level by level according to the  $C_i^d (i \in [1, K])$ . An example of frequency bands calculation of the first three levels is given to illustrate this operation, as shown in Fig. 5. The node  $D_0$  expresses the original signal, and the  $D_1^0$  as well as the  $D_1^1$  are the nodes of level 1. The red line in Fig. 5(a) is the longest SSC  $C_1^d$ , and its corresponding frequency is  $f_1$ .

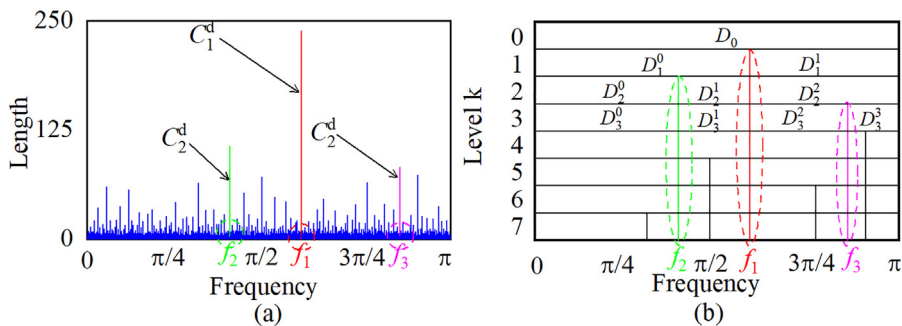


Fig. 5. Illustration of paving (frequency/frequency resolution) of the plane adaptive generation.

Then the frequency bands of  $D_1^0$  and  $D_1^1$  can be separated by  $C_1^d$ , as the red line shown in Fig. 5(b), and expressed as  $[0, f_1]$  and  $[f_1, \pi]$  (normalized frequency  $2\pi$ ). In Level 2, the two longest SSCs are utilized to obtain the frequency bands. Since the locations of SSCs in frequency domain are inconsistent, the frequencies should be sorted in a descending order before generating the boundaries. As shown in Fig. 5(a), the second long SSC (green line) is  $C_2^d$  and its corresponding frequency is  $f_2$ . The frequency bands of nodes ( $D_2^m(m \in [0, 2])$ ) in level 2 are divided by  $C_1^d$  (red line) and  $C_2^d$  (green line), as shown in Fig. 5(b). The frequency bands of level 2 are equal to  $[0, f_2]$ ,  $[f_2, f_1]$  and  $[f_1, \pi]$ , respectively. Similarly, the frequency bands of level 3 can be calculated according to the three longest SSCs (red line, green and purple line), and the frequency bands of level 3 are  $[0, f_2]$ ,  $[f_2, f_1]$ ,  $[f_1, f_3]$  and  $[f_3, \pi]$ . Repeat the above steps to automatically complete band segmentation.

As introduced above, the frequency band generation is achieved by the SSCs, and the SSC is utilized in the following steps. The proposed method takes  $K$  SSCs to obtain the frequency bands in level  $K$ , and the number of the frequency bands is equal to the  $K$ . With an increase of the decomposition level, the Fourier spectrum is separated into a series of frequency bands, which will result in the frequency band that is too narrow. Narrow band will lose the fault-related information. Meanwhile, as the decomposition levels increases, the computational cost inevitably increases as well. In this article, the maximum decomposition level is set to 16.

### 3.2. The empirical wavelet transform method

The frequency boundary of each node is automatically obtained according to the aforementioned step, and the first shortcoming is eliminated. Concurrently, the EWT is utilized to further refine the proposed method. As mentioned above, the robustness properties of the kurtosis criterion are heavily affected by the signal length. To reduce the influence of the signal length, the decomposition of the narrow-band signal is accomplished by the EWT method. This step aims to guarantee that the length of the filtered signal is consistent. The EWT can adaptively construct a series of filters in the frequency domain according to the detected boundaries, and can ensure that the Fourier spectrum of each mode is of compact support. Assume that we want to decompose the signal into  $N$  modes, we need to define  $N-1$  boundaries in the Fourier domain (with normalized frequency). Each segment is denoted as  $\Delta_n = [\omega_{n-1}, \omega_n]$  ( $n = 1, 2, \dots, N$ ), where  $\omega_0 = 0$ ,  $\omega_N = \pi$  and  $\bigcup_{n=1}^N \Delta_n = [0, \pi]$ . The EWT can be treated as the application of an adaptive filter bank which means that each filter corresponds to one Fourier support. The filters are constructed, which is inspired by the idea utilized in deriving Meyer's wavelet, and the empirical wavelets  $\psi_n(\omega)$  and empirical scaling function  $\phi_n(\omega)$  are expressed as follows, respectively.

$$\phi_n(\omega) = \begin{cases} 1 & \text{if } |\omega| \leq (1 - \gamma)\omega_n \\ \cos\left[\frac{\pi}{2}\eta\left(\frac{1}{2\tau_n}(|\omega| - \omega_n + \tau_n)\right)\right] & \text{if } (1 - \gamma)\omega_n \leq |\omega| \leq (1 + \gamma)\omega_n \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

$$\psi_n(\omega) = \begin{cases} 1 & \text{if } (1 + \gamma)\omega_n \leq |\omega| \leq (1 - \gamma)\omega_{n+1} \\ \cos\left[\frac{\pi}{2}\eta\left(\frac{1}{2\tau_{n+1}}(|\omega| - \omega_{n+1} + \tau_{n+1})\right)\right] & \text{if } (1 - \gamma)\omega_{n+1} \leq |\omega| \leq (1 + \gamma)\omega_{n+1} \\ \sin\left[\frac{\pi}{2}\eta\left(\frac{1}{2\tau_n}(|\omega| - \omega_n + \tau_n)\right)\right] & \text{if } (1 - \gamma)\omega_n \leq |\omega| \leq (1 + \gamma)\omega_n \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where  $\eta(x) = x^4(35 - 84x + 70x^2 - 20x^3)$ ;  $\gamma$  is the parameter related to the width of the transition phase  $T_n$ .

The width of the transition phase  $T_n$  is defined as  $2\tau_n$ , and the simplest choice of  $\tau_n$  is  $\tau_n = \gamma\omega_n$  ( $0 < \gamma < 1$ ). Concurrently,  $\gamma < \min_n \left( \frac{\omega_{n+1} - \omega_n}{\omega_{n+1} + \omega_n} \right)$  is adopted to guarantee the EWT to be a tight frame.

After that, the signal decomposition will be achieved by the inner product of the signal with the obtained empirical wavelets and empirical scaling function. Thus, the approximation coefficient  $W_x^e(0, t)$  and detail coefficients  $W_x^e(n, t)$  of the signal  $x(n)$  can be expressed as follows:

$$W_x^e(0, t) = \langle x(n), \phi_1(n) \rangle = \int x(\tau) \overline{\phi_1(\tau - t)} d\tau = \text{IFT}(\text{FT}(x(\tau)) \times \overline{\phi_1(\omega)}) \quad (6)$$

$$W_x^e(n, t) = \langle x(n), \psi_n(n) \rangle = \int x(\tau) \overline{\psi_n(\tau - t)} d\tau = \text{IFT}(\text{FT}(x(\tau)) \times \overline{\psi_n(\omega)}) \quad (7)$$

where the FT denotes the Fourier transform and the IFT denotes the inverse Fourier transform;  $\langle \cdot, \cdot \rangle$  is the inner product; and  $\overline{(\cdot)}$  is the complex conjugate operator.

Obviously, the EWT can realize the signal adaptive decompositions by the predetermined segments of Fourier spectra, and these frequency boundaries have been automatically identified in Section 3.1.

### 3.3. Correlated kurtosis criterion

Following the flowchart of Kurtogram generation, another core step is to obtain a two-dimension map which clearly reveals the presence of resonant frequency bands and makes easy identification of the fault characteristic frequencies in the envelope spectrum. Thus, the effectiveness of the Kurtogram is inextricably linked to the reliability of the selected indicator. As mentioned above, the collected axle-box vibration signals can be generally categorized into two impulsive components: the faulty bearing periodic impulses and repetitive transients induced by wheelset-tread defects. Therefore, the impulsive and the periodic properties should be considered simultaneously. For that purpose, the correlated kurtosis (CK) is employed to indicate the frequency bands of the periodic impulses with the specified period  $T$ . The first shift correlated kurtosis  $CK_1(T)$  of period  $T$  is defined as [32]:

$$CK_1(T) = \frac{\sum_{n=1}^N (y_n y_{n-T})^2}{\left(\sum_{n=1}^N y_n^2\right)^2} \quad (8)$$

where  $y_n$  is the complex envelope of the filtered signal, and  $N$  is the length of the filtered signal.

Meanwhile,  $M$ -shift correlated kurtosis  $CK_M(T)$  of period  $T$  is proposed to improve the sensitivity to the periodic impulses. The  $M$ -shift correlated kurtosis  $CK_M(T)$  can be expressed as:

$$CK_M(T) = \frac{\sum_{n=1}^N \left(\prod_{m=0}^M y_{n-mT}\right)^2}{\left(\sum_{n=1}^N y_n^2\right)^{M+1}} \quad (9)$$

where  $M$  is the shift order and should be a positive integer. It should be noted that with the order shift  $M$ 's increase, the fault period  $T$  should be better estimated [32].

The goal of the CK indicator is to select the fault-related sub-band signals by taking the impulsiveness and periodicity into consideration. When the acquired signal contains several periodic impulses, the traditional indicators, such as kurtosis and envelope spectrum kurtosis, may become invalid. Because the correlated kurtosis has a strong power to characterize the repetitive transients with an artificial set period  $T$ , it is particularly valuable to process the multiple source vibration signals, for instance, the wheelset-bearing systems' vibration signals. In addition, the CK is not only immune to random impulses by taking the period  $T$  into consideration, but also immune to harmonics interfaces by calculating the time-domain kurtosis.

## 4. Simulation study

To illustrate the effectiveness of the proposed method, a simulated signal  $y(t)$  which mimics the vibrations of a wheelset-bearing system [34] is expressed as follows:

$$y(t) = y_b(t) + y_w(t) + y_h(t) + y_i(t) + N(t) \quad (10)$$

where  $y_b(t)$  denotes the impulse response signal induced by bearing outer race fault with the fault characteristic frequency  $f_{BPFO}$ ;  $y_w(t)$  is the impulse response signal caused by wheelset-tread defect with wheelset rotation frequency  $f_r$ ;  $y_h(t)$  is the harmonic interference;  $y_i(t)$  is the random impulse response signal;  $N(t)$  is the Gaussian white noise at a  $-5$  dB SNR of  $y_b(t)$ .

$y_b(t)$ ,  $y_w(t)$ ,  $y_i(t)$  and  $y_h(t)$  are respectively expressed as follows:

$$y_b(t) = \sum_{n=0}^{N_b} L_b \exp(-\beta_b(t - nT_b) \cos(2\pi\omega_b(t - nT_b - \tau_b))) u(t - nT_b - \tau_b) \quad (11)$$

$$y_w(t) = \sum_{n=0}^{N_w} L_w \exp(-\beta_w(t - nT_w) \cos(2\pi\omega_w(t - nT_w - \tau_w))) u(t - nT_w - \tau_w) \quad (12)$$

$$y_i(t) = A_i e^{-\beta_i t} \cos(\omega_i t) \quad (13)$$

$$y_h(t) = 0.4 \sin(2\pi f_r t) + 0.2 \sin(4\pi f_r t) + 0.2 \sin(6\pi f_r t) \quad (14)$$

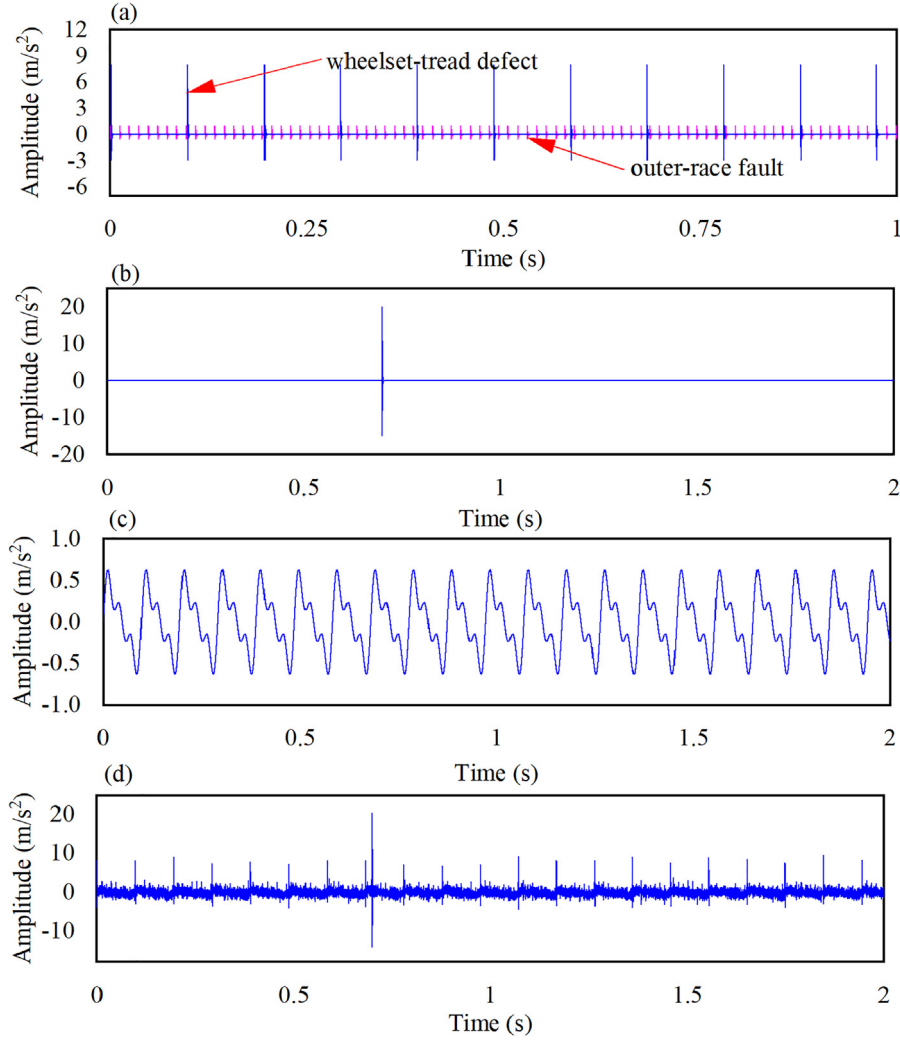
where  $L_b = \sqrt{A_b^2(\beta_b^2 - \omega_b^2)^2 + 4A_b^2\beta_b^2\omega_b^2}$  and  $L_w = \sqrt{A_w^2(\beta_w^2 - \omega_w^2)^2 + 4A_w^2\beta_w^2\omega_w^2}$ ;  $A_b$ ,  $A_w$  and  $A_i$  are the amplitudes of the impulse responses;  $\beta_b$ ,  $\beta_w$  and  $\beta_i$  are the structure damper coefficient;  $\omega_b$ ,  $\omega_w$  and  $\omega_i$  are the excited resonance frequencies;  $N_b$  and  $N_w$  are the number of impulses;  $T_b = f_{BPFO}^{-1}$ ,  $T_w = f_r^{-1}$ ;  $\tau_b$  and  $\tau_w$  are the random slippages. The details of the simulated signal are listed in Table 1. The sampling frequency is 10 kHz, and the length of the signal is 2 s.

The simulated signal is shown in Fig. 6. Obviously, the simulated signal is of complexity, which contains several interferences such as the harmonic interference and the random impulse interference. These interferences lead to a major challenge for the rotation machine diagnosis. The Fourier spectrum and envelope spectrum are depicted in Fig. 7(a) and (b). As shown in Fig. 7(b), the primary frequencies in the envelope spectrum are the wheelset rotation frequency  $f_r$  and its harmonics.

**Table 1**

Parameters for the simulated signal expressed by Eq. (10).

| $k$ | $A_k$ | $\beta_k$ (Ns/m) | $\omega_k$ (Hz) | $T_k$ (s) | $\tau_k$    |
|-----|-------|------------------|-----------------|-----------|-------------|
| $b$ | 1     | 1500             | 3500            | 1/83.3    | $0.01 T_b$  |
| $w$ | 8     | 2000             | 1000            | 1/10.29   | $0.005 T_w$ |
| $i$ | 20    | 1500             | 2500            | –         |             |

**Fig. 6.** The simulation signal: (a) impulse responses induced by bearing fault and wheelset-tread defect; (b) random impulse; (c) harmonic interference; (d) the total simulation signal.

Firstly, the proposed ACK is used to analyze the simulated signal. According to the flowchart of the proposed method, the SSP is obtained through the SSR method, and depicted in Fig. 8(a). The maximum level is set to 16. The frequency boundary of each node is adaptively generated in accordance with the SSP, and the paving of the plane is illustrated in Fig. 8(b). It is worth noting that the Fourier supports of the nodes are divided unevenly and achieved by the SSP. After that, the correlated kurtosis value with  $T = 1/10.29$  s (related to the wheelset rotation frequency  $f_r$ ),  $M = 4$  of each node is calculated and filled into the paving of the plane for the generation of the adaptive correlated Kurtogram. Fig. 9(a) gives the adaptive correlated Kurtogram and the maximum correlated kurtosis value appears at Fourier support [146.5 Hz, 1572 Hz], and the envelope and envelope spectrum are displayed in Fig. 9(b) and (c), respectively. From Fig. 9(c), the wheelset rotation frequency  $f_r$  and its harmonics can be clearly identified as the red dots indicate. The adaptive correlated Kurtogram with  $T = 1/83.3$  s (related to the bearing outer race fault with the fault characteristic frequency  $f_{BPFO}$ ),  $M = 4$  is generated and shown in Fig. 9(d). The node with Fourier support [3217.5 Hz; 3940 Hz] has the highest kurtosis value as shown in Fig. 9(d), and the envelope and envel-

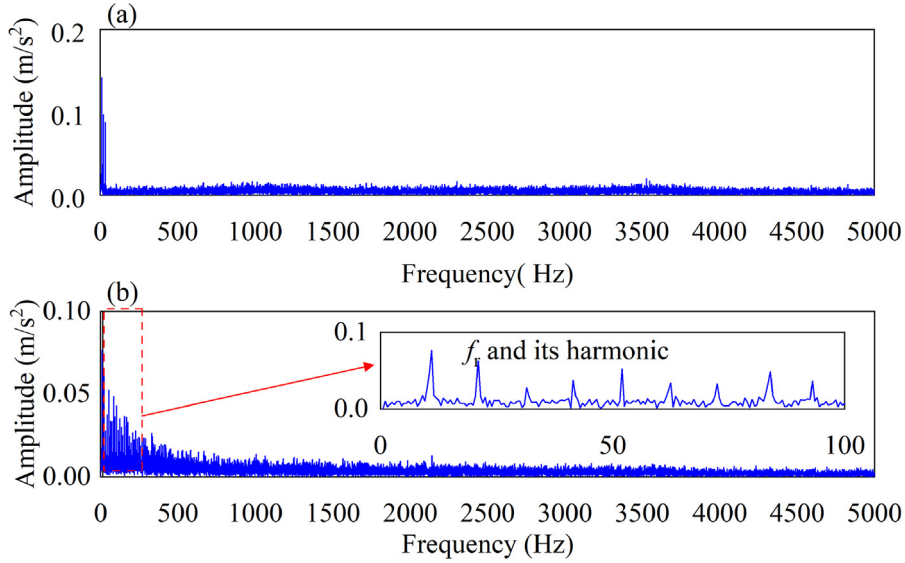


Fig. 7. (a) the Fourier spectrum and (b) the envelope spectrum of the simulated signal.

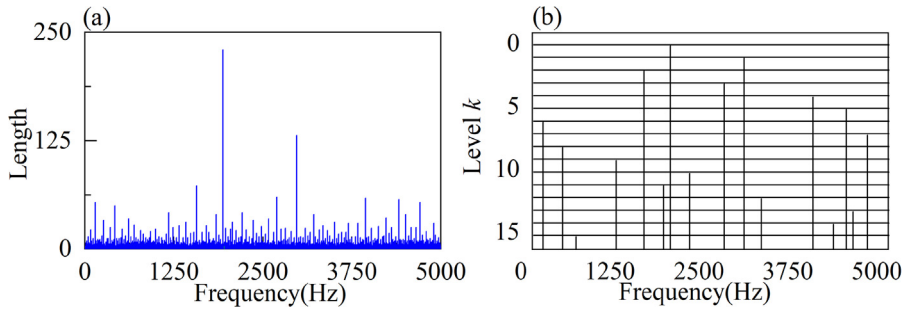


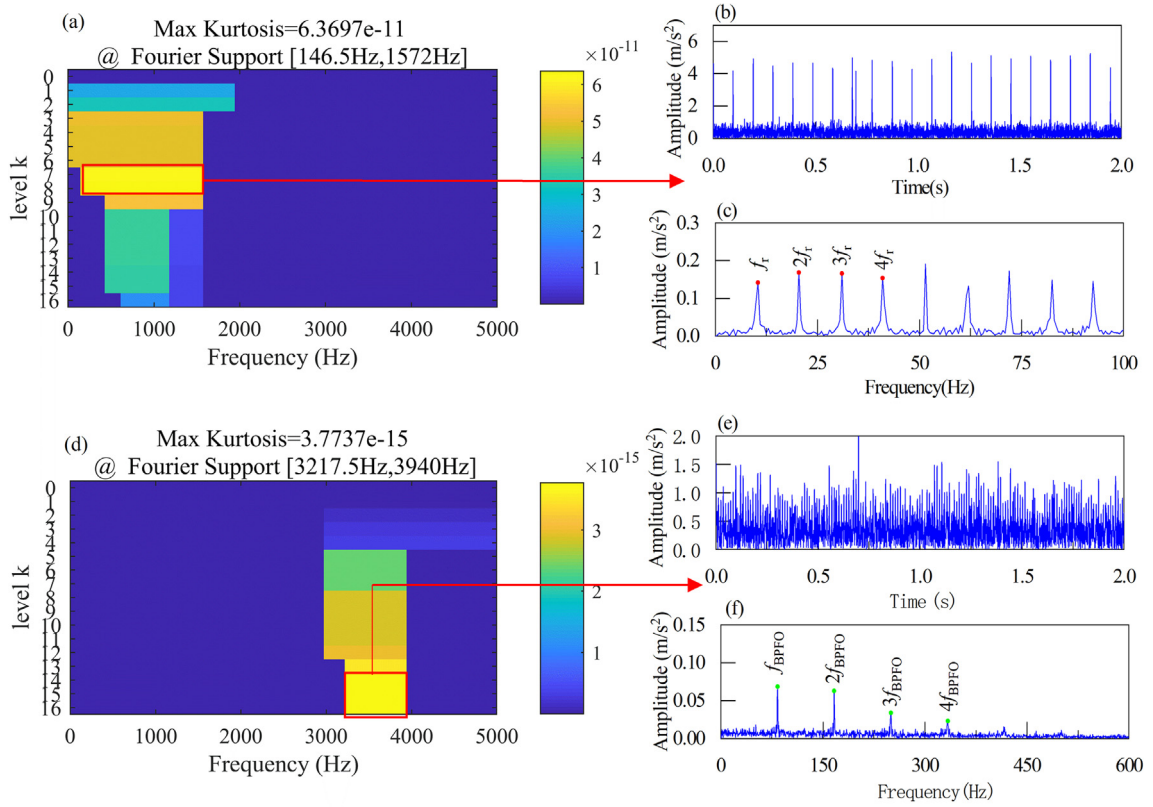
Fig. 8. (a) the SSP of the simulated signal and (b) the paving of the plane of (a).

ope spectrum of this node are displayed in Fig. 9(e) and (f), respectively. The  $f_{BPFO}$  and its harmonics can be recognized in Fig. 9(f) as the green dots indicate. From the above results, we can see that the wheelset-bearing system's fault diagnosis (including the wheelset-tread defect and bearing fault) can be achieved by the proposed ACK method.

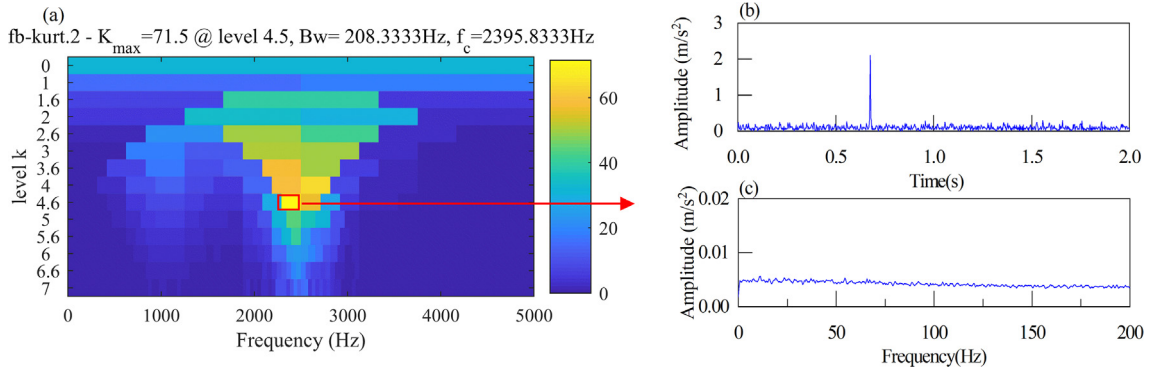
Furthermore, the FK is applied to make the Kurogram, and the result is shown in Fig. 10(a). The node with the highest kurtosis value is located at level 4.5 with center frequency 2395.8333 Hz and bandwidth 208.3333 Hz. The envelope and envelope spectrum of the filtered signal are displayed in Fig. 10(b) and (c). Obviously, none of the useful frequency information can be detected from the envelope spectrum Fig. 10(c) due to the extracted component is the random impulse as shown in Fig. 10(b). Therefore, the FK is inappropriate to realize repetitive transients' extraction when the signal contains the random impulse because of the kurtosis indicator is sensitive to the random impulse.

Meanwhile, the Autogram is adopted to process the simulated signal, and the Autogram is shown in Fig. 11(a). The maximum kurtosis is found with center frequency 1562.5 Hz and bandwidth 625 Hz at level 3. The envelope spectrum of the filtered signal is presented in Fig. 11(b). The wheelset rotation frequency  $f_r$  and its harmonics are revealed in the envelope spectrum, which demonstrates that the resonant frequency of the wheelset-tread defect has been detected. However, the bearing fault-related information cannot be extracted.

A further discussion is conducted to analyze the results obtained by the three methods. The detected boundaries are shown in Fig. 12. The FK uses the kurtosis indicator to express the impulsiveness of the sub-band signals. When the analyzed signal contains the random impulses, the FK may obtain an error result due to the kurtosis indicator is sensitive to the random impulses as the green lines indicate. The Autogram utilizes the kurtosis of the unbiased Autocorrelation (AC) instead of the kurtosis to find the desired frequency band which contains fruitful fault-related information. The result indicates that the Autogram is a powerful method for the extraction of repetitive transients, especially the signal where the heavy interferences exist, such as the harmonic interference and the random impulse interference. However, the wheelset-bearing system vibration signal contains two types of repetitive transient components, and Autogram is invalid in the identification of the specific periodic impulses from several repetitive transient components. Therefore, the red lines indicate that the Autogram



**Fig. 9.** The results of the simulated signal of the proposed ACK method: (a) the correlated Kurtogram with  $T = 1/10.29$  s,  $M = 4$ ; (b) the envelope and (c) the envelope spectrum of the filtered signal with Fourier support [146.5 Hz, 1572 Hz]; (d) correlated Kurtogram with  $T = 1/83.3$  s,  $M = 4$ ; (e) the envelope and (f) envelope spectrum of the filtered signal with Fourier support [3217.5 Hz, 3940 Hz].



**Fig. 10.** (a) the Kurtogram of the simulated signal obtained by FK; (b) the envelope of the filtered signal; (c) the envelope spectrum of the filtered signal.

can only realize wheelset-tread defect detection, but fails to identify the resonant frequency band of bearing fault. Additionally, Autogram adopts a fixed resolution method to generate sub-band signals and so that it can only identify the right part of the actual resonant frequency bands (the identified frequency band misses the resonant frequency 1000 Hz). It is evident that neither the FK nor the Autogram can correctly and rationally detect the two expected frequency bands. The proposed ACK method utilizes the correlated kurtosis as the criterion to represent the impulsive symptoms, and achieves resonant frequency bands identification with a specific periodic. The correlated kurtosis possesses the property of highlighting the repetitive transients with a specific periodic, and is ideal for diagnosis of the wheelset-bearing system's signal. The purple lines indicate that the proposed ACK has successfully identified the resonant frequency bands of the two parts. Compared with the Autogram result, the ACK method selects the entire frequency band which contains most energy-rich information by its adaptive property, and it is beneficial to reveal fault characteristic frequency in the envelope spectrum.

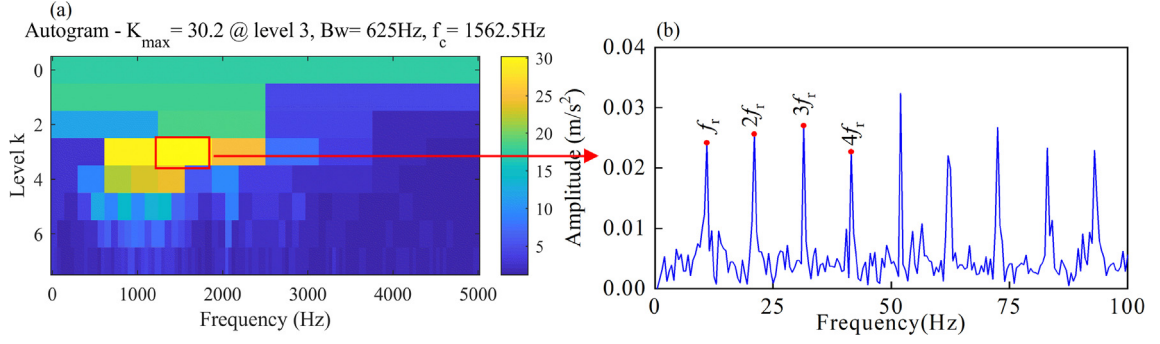


Fig. 11. (a) the Autogram of the simulated signal and (b) the envelope spectrum of the filtered signal.

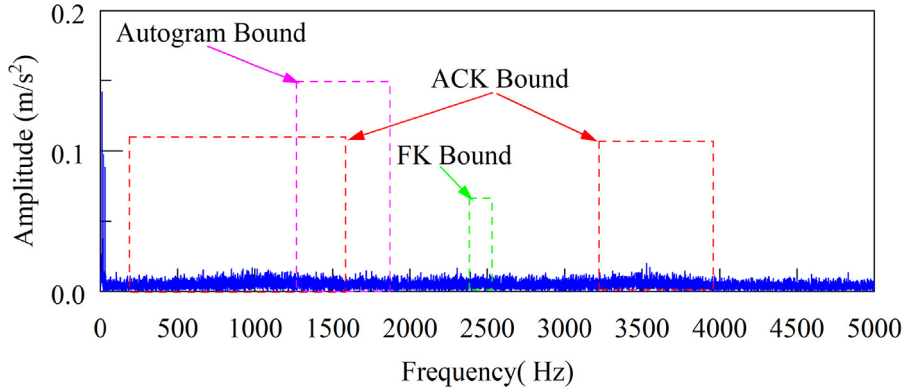


Fig. 12. The boundaries of three methods.

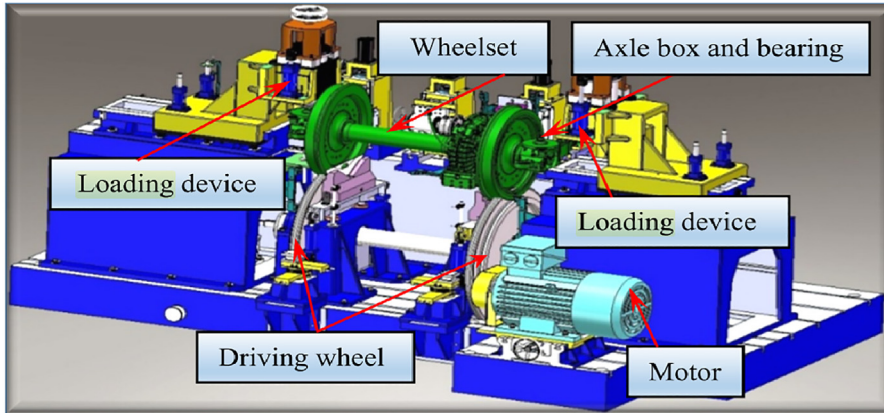
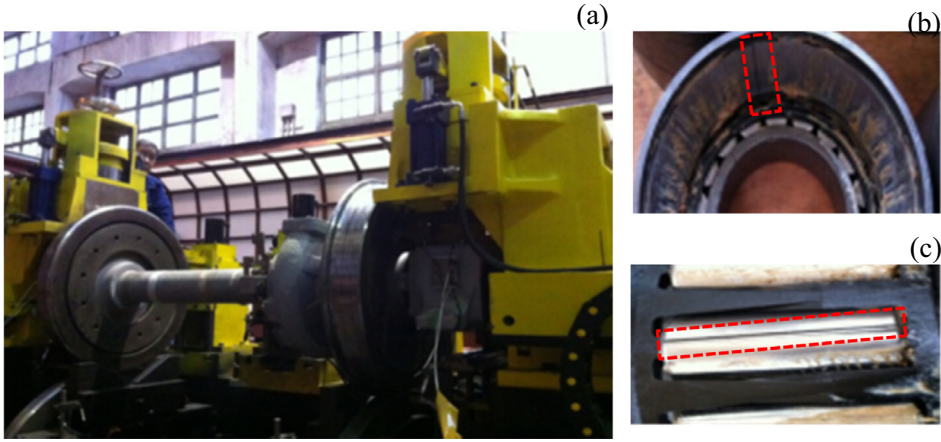


Fig. 13. The structure of the test rig.

## 5. Experiments validation

In this section, the vibration signals collected from a wheelset-bearing test bench are utilized to validate the effectiveness of the proposed ACK method. The experiments are conducted in the State Key Laboratory of Traction Power of the Southwest Jiaotong University. The structure of the test rig is shown in Fig. 13. The test rig includes an entire wheelset-bearing system and a driving subsystem. The driving subsystem contains a motor and a driving wheel. The driving wheel transmits the traction power to the wheelset. The test rig uses a track wheel with a finite radius to replace the actual straight track. Hereafter, the wheel-to-wheel contact can be approximated to the wheel-to-rail contact in the real line-service environment, and can ensure that the collected signal has a similar form to the axle-box vibration signal acquired in a practical running condition.

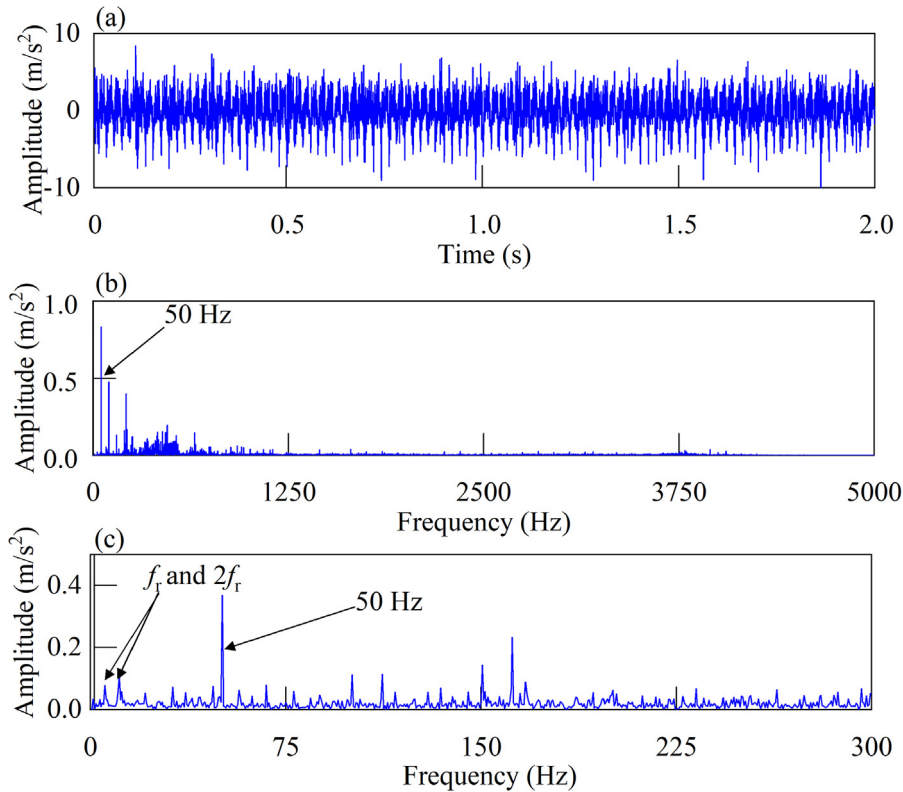


**Fig. 14.** The test rig and two artificial bearing defects: (a) test rig, (b) outer race defect, and (c) roller defect.

of the train. The experiments can be categorized into two cases: in case 1, the single outer race defect is considered; in case 2, outer race defect and roller defect are concurrently considered. Meanwhile, the wheelset is disassembled from an in-service train, and some little defects exist in the wheelset-tread, which will cause the repetitive transients. The test rig is plotted in Fig. 14(a) and two kinds of artificial defects on the outer race and rollers are shown in Fig. 14(b) and Fig. 14(c), respectively. The parameters of the tested bearing are listed in Table 1.

The outer-race fault characteristic frequency  $f_{BPFO}$ , the rolling fault characteristic frequency  $f_{BSF}$ , and the fundamental cage frequency  $f_c$  are expressed as:

$$f_{BPFO} = \frac{Zf_r}{2} \left( 1 - \frac{d}{D} \cos \alpha \right) \quad (15)$$



**Fig. 15.** The acquired vibration signal with bearing outer race defect: (a) time waveform; (b) Fourier spectrum; (c) envelope spectrum.

$$f_{BSF} = \frac{Df_r}{2d} \left(1 - \left(\frac{d}{D} \cos \alpha\right)^2\right) \quad (16)$$

$$f_c = \frac{f_r}{2} \left(1 - \frac{d}{D} \cos \alpha\right) \quad (17)$$

where  $f_r$  is the wheelset rotation frequency of the wheelset;  $d$  is the roller diameter;  $D$  denotes the pitch diameter;  $Z$  is the number of rollers; and  $\alpha$  denotes the contact angle.

### 5.1. Case 1: Bearing with outer race fault

In the first case, the bearing has suffered an outer race fault. The sampling frequency  $f_s$  is 10 kHz, and the signal length is 2 s. The rotation frequency of the wheelset  $f_r$  is set to 5.1 Hz. According to the Eq. (15) and Table 1, the outer-race fault char-

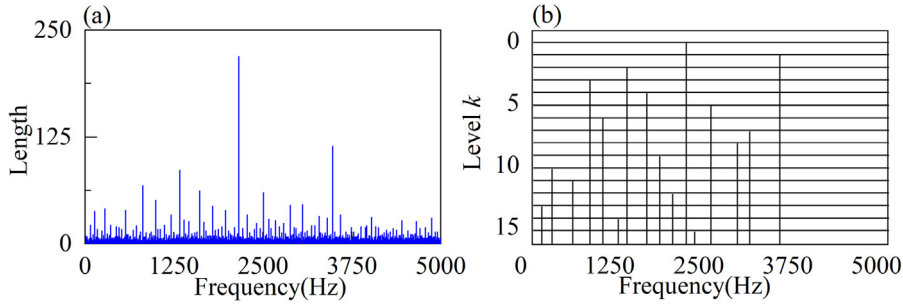


Fig. 16. (a) the SSP of the bearing outer race fault signal and (b) the paving of the plane of (a).

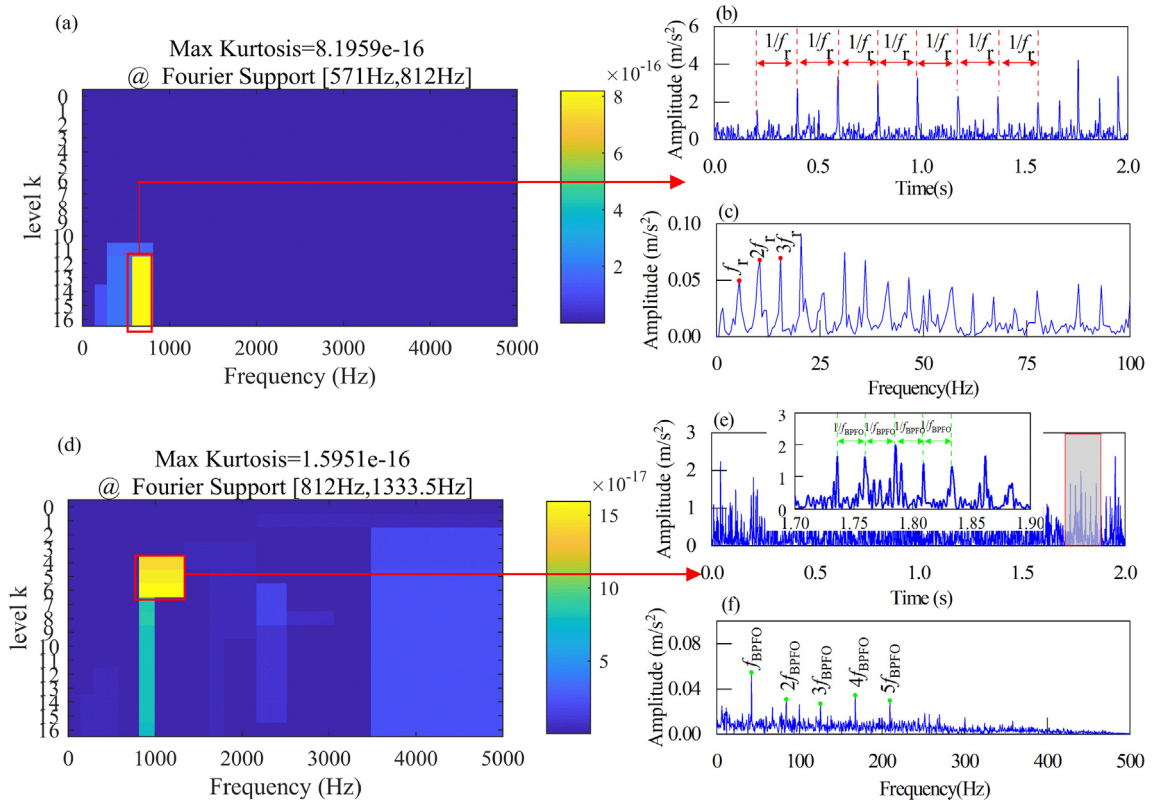


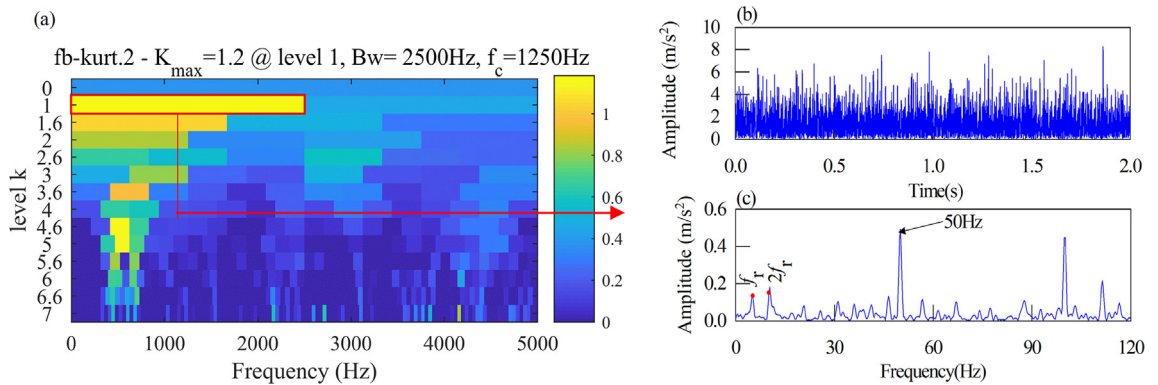
Fig. 17. The results of the bearing outer race defect signal of the proposed ACK method; (a) the correlated Kurtogram with  $T = 1/5.1$  s,  $M = 4$ ; (b) the envelope and (c) the envelope spectrum of the filtered signal with Fourier support [517 Hz, 812 Hz]; (d) the correlated Kurtogram with  $T = 1/41.6$  s,  $M = 4$ ; (e) the envelope and (f) the envelope spectrum of the filtered signal with Fourier support [812 Hz, 1333.5 Hz].

acteristic frequency  $f_{BPFO}$  is equal to 41.6 Hz. Fig. 15(a) shows the time waveform of the collected signal, and the Fourier spectrum and envelope spectrum are presented in Fig. 15(b) and (c), respectively. Obviously, a strong power frequency exists in the Fourier spectrum and envelope spectrum as shown in Fig. 15(b) and (c), and the rotation frequency and fault characteristic frequency can be hardly detected.

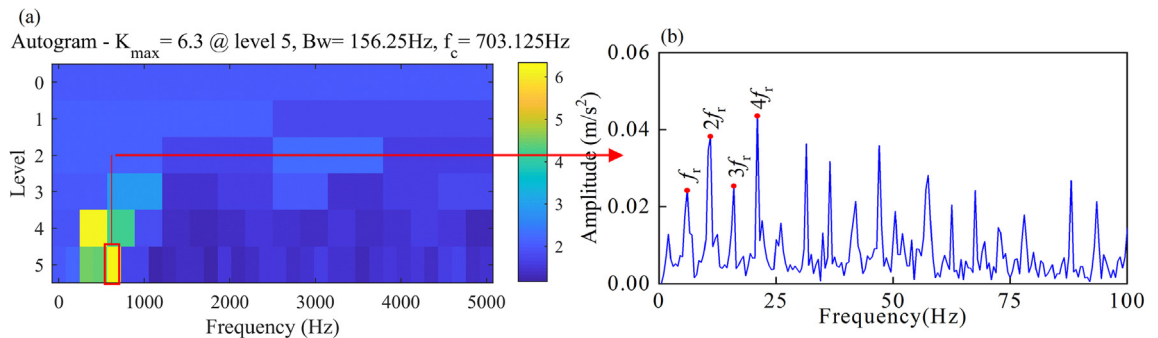
Firstly, the proposed method is utilized to process the collected signal. According to the flowchart of the proposed method, the SSP is firstly generated and plotted in Fig. 16(a). Hereafter, the paving of the plane is automatically obtained and presented in Fig. 16(b). The correlated kurtosis (with parameters  $T=1/f_r = 1/5.1$  s and  $M = 4$ ) of each node is calculated and filled into Fig. 16(b) to obtain the adaptive correlated Kurtogram, the result is shown in Fig. 17(a). The result shows that the maximum correlated kurtosis value is located at the Fourier support [571 Hz; 812 Hz]. The envelope and envelope spectrum of the highlighted frequency band are depicted in Fig. 17(b) and (c), respectively. In Fig. 17(b), the impulses with period  $1/f_r$  can be detected. Meanwhile, in Fig. 17(c), the most active components are wheelset-rotation frequency  $f_r$  and its harmonics, as indicated by red dots. After that, the adaptive correlated Kurtogram with parameters  $T = 1/f_{BPFO} = 1/41.6$  s and  $M = 4$  is calculated and expressed in Fig. 17(d). The largest correlated kurtosis value appears at the frequency band [812 Hz, 1333.5 Hz]. Using EWT to extract component, the envelope and envelope spectrum are shown in Fig. 17(e) and (f), respectively. It can be found that there are several bearing outer race fault-related impulses expressed in the envelope of the filtered signal. Meanwhile, the bearing outer race fault characteristic frequencies  $f_{BPFO}$  and its harmonics can be detected in the envelope spectrum, which indicates the bearing has suffered an outer race defect. We can conclude that the proposed method has a strong ability to realize wheelset-bearing system's diagnosis.

Meanwhile, the FK method is employed to analyze the collected signal, and the Kurtogram is shown in Fig. 18(a). The highest value appears in the frequency band with center frequency 1250 Hz, bandwidth 2500 Hz, and the envelope and envelope spectrum of the filtered signal are illustrated in Fig. 18(b) and (c), respectively. Similar to the envelope spectrum of the original signal, the fundamental frequency components in the envelope spectrum are the power interference frequencies, and the amplitudes of the rotation frequency and its harmonics are small, as the red dots indicate. None of the bearing faulty information can be detected from the envelope spectrum. Thus, the FK fails to detect the desired frequency band. It is hard to decide whether the out-race fault exists in the bearing.

Autogram is utilized to process the collected signal, and the result is depicted in Fig. 19(a). The node in level 5 with center frequency 703.125 Hz and bandwidth 156.25 Hz has the largest kurtosis 6.3. After extracting the narrow-band signal and



**Fig. 18.** (a) the Kurtogram of the hearing outer race defect signal obtained by the FK; (b) the envelope of the filtered signal; (c) the envelope spectrum of the filtered signal.



**Fig. 19.** (a) the Autogram of the bearing outer race defect signal and (b) the envelope spectrum of the filtered signal.

**Table 2**

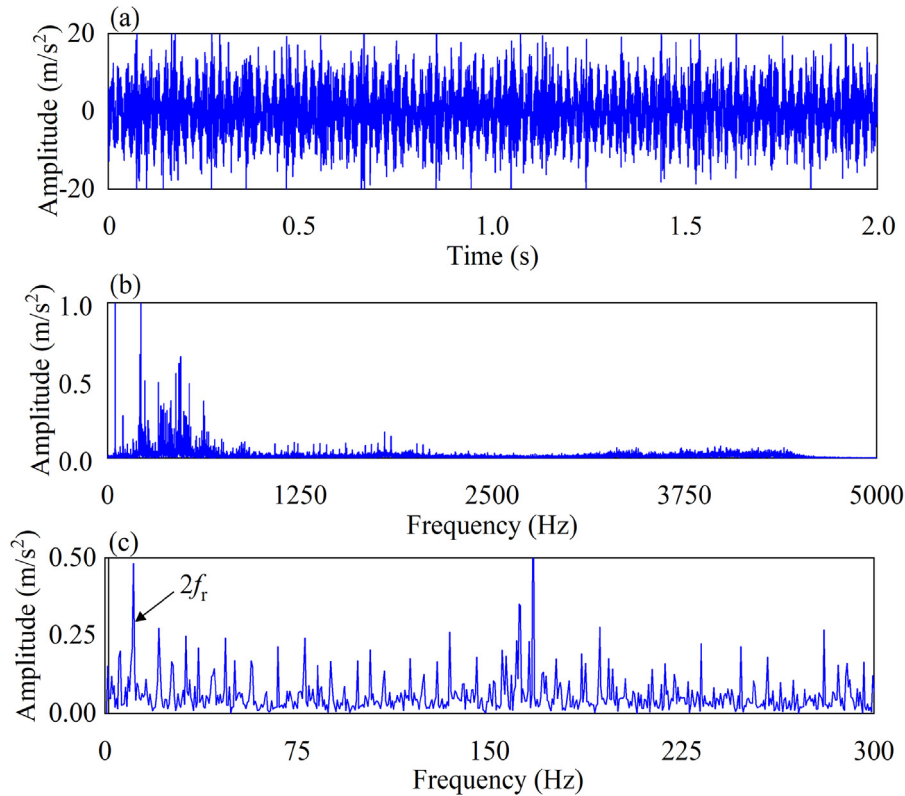
The parameters of the tested wheelset bearing.

| Roller number | Roller diameter (mm) | Pitch diameter (mm) | Contact angle (rad) | Load (kN) |
|---------------|----------------------|---------------------|---------------------|-----------|
| 19            | 26.9                 | 180                 | 0.1571              | 50        |

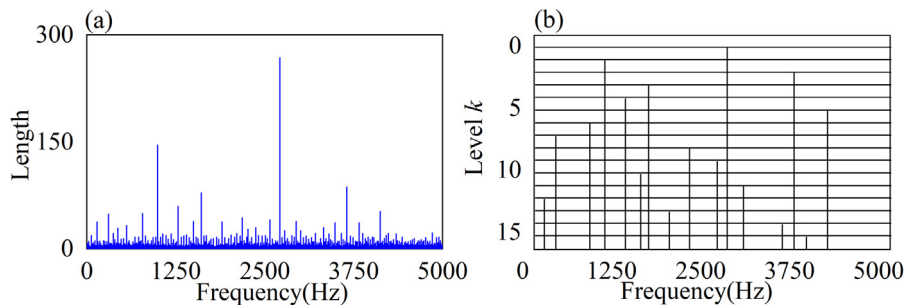
conducting envelope spectrum analysis, the result is presented in the Fig. 19(b). As shown in the figure, the rotation frequency  $f_r$  and its harmonics are obviously identified in the Fig. 19(b), as the red dots indicate. However, the bearing fault-related information is overwhelmed by interference, and neglected in the envelope spectrum. It is demonstrated that this frequency region is related to the wheel/wheel excitation and unrelated to the faulty bearing, and the bearing fault detection failed.

### 5.2. Case 2: Bearing with compound faults

In the second case, the bearing has suffered multiple faults, including an outer race defect and a rolling element defect. Similar to case 1, the wheelset is also disassembled from an in-service train. The shaft rotation frequency  $f_r$  of wheelset is



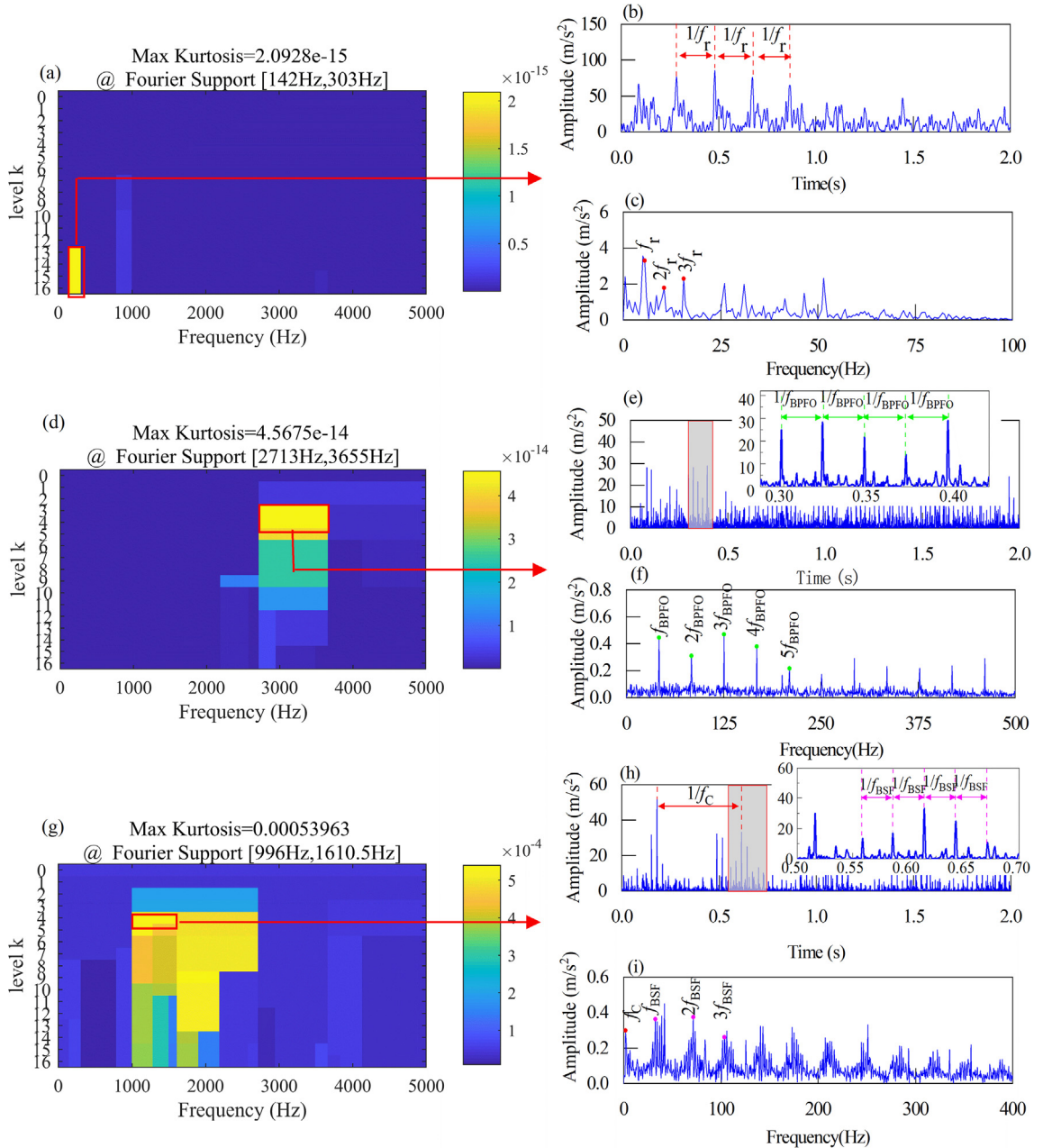
**Fig. 20.** The acquired vibration signal with bearing multiple defects: (a) time waveform; (b) Fourier spectrum; (c) envelope spectrum.



**Fig. 21.** (a) the SSP of the bearing multiple defects signal and (b) the paving of the plane of (a).

5.1 Hz. According to the Eq. (15) and Table 2, the rolling element fault characteristic frequency  $f_{BSF}$  is equal to 33.9 Hz, and the fundamental cage frequency  $f_c = 2.19$  Hz. Fig. 20(a) shows the time waveform of the collected signal, and the Fourier spectrum and envelope spectrum of the acquired signal are depicted in Fig. 20(b) and (c), respectively. As shown in Fig. 20(c), the primary frequency in the envelope spectrum is  $2f_r$ , and the bearing fault-related frequencies are weakly exposed.

The proposed ACK method is utilized to process the collected signal. After using the SSR method to obtain the SSP the result is illustrated in Fig. 21(a). The paving of the plane is automatically generated, and depicted in the Fig. 21(b). Thereafter, the correlated kurtosis values with different parameters are calculated and filled into the Fig. 21(b). The Fig. 22(a) illustrates the adaptive correlated Kurtogram with  $T=1/f_r = 1/5.1$  s,  $M = 4$ , which is corresponding to the rotation frequency. The highest correlated kurtosis appears in the node with the Fourier support [142 Hz, 303 Hz]. A series of repetitive transients with inter-

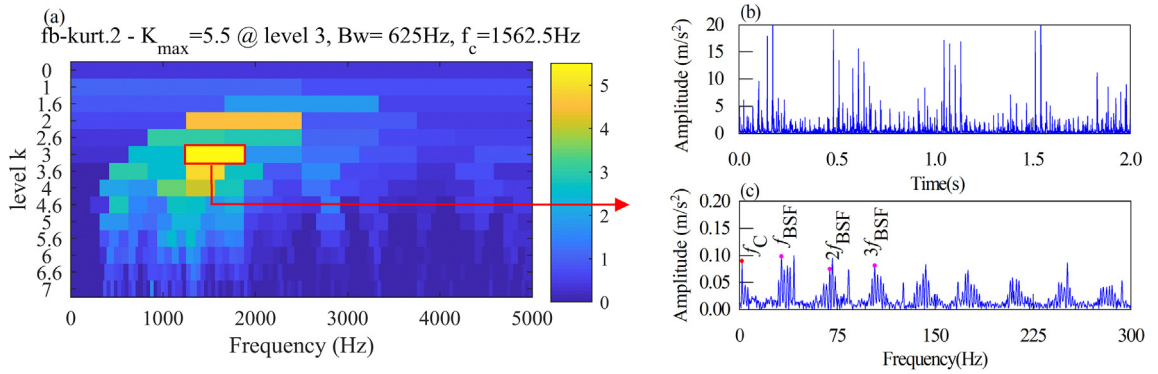


**Fig. 22.** The results of the bearing multiple defects signal of the proposed ACK method: (a) the correlated Kurtogram with  $T = 1/5.1$  s,  $M = 4$ ; (b) the envelope and (c) the envelope spectrum of the filtered signal with the Fourier support [142 Hz, 303 Hz]; (d) the correlated Kurtogram with  $T = 1/41.6$  s,  $M = 4$ ; (e) the envelope and (f) the envelope spectrum of the filtered signal with the Fourier support [2713 Hz, 3655 Hz]; (g) the correlated Kurtogram with  $T = 1/33.9$  s,  $M = 1$ ; (h) the envelope and (i) the envelope spectrum of the filtered signal with the Fourier support [996 Hz, 1610.5 Hz].

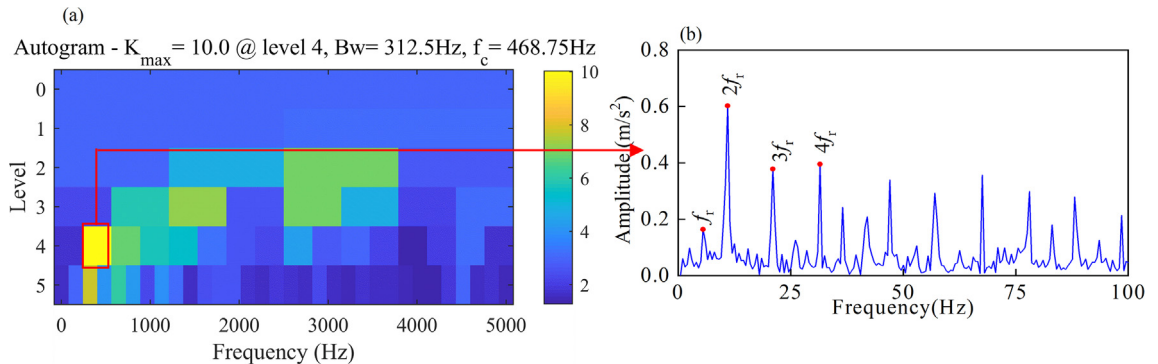
vals  $f_r$  can be observed in the envelope of the filtered signal as shown in the Fig. 22(b). Meanwhile, the  $f_r$  and its harmonics are marked by red dots in the envelope spectrum Fig. 22(c). Subsequently, the Fig. 22(d) shows the adaptive correlated Kurtogram with  $T=1/f_{BPFO} = 1/41.6$  s,  $M = 4$ , which is corresponding to the bearing outer race fault characteristic frequency. The highest correlated kurtosis appears in the node with the Fourier support [2713 Hz, 3655 Hz]. The Fig. 22(e) and the Fig. 22(f) are the envelope and envelope spectrum of the filtered signal. From the zoom-in view in the envelope, we can find stronger impulses with the distance  $1/f_{BPFO}$ , and the dominant frequencies in the envelope spectrum are  $f_{BPFO}$  and its harmonics. The Fig. 22(g) depicts the adaptive correlated Kurtogram with  $T=1/f_{BSF} = 1/33.9$  s,  $M = 1$ , which is corresponding to the bearing rolling element fault. Due to the damaged portion of the rolling element moving in-and-out of the load zone, the amplitudes of the impulses are modulated by the cage frequency. Therefore, a smaller shift order  $M$  is adopted to reduce the influence of the amplitude divergence. The highest correlated kurtosis appears in the node with the Fourier support [996 Hz, 1610.5 Hz]. The Fig. 22(h) and the Fig. 22(i) are the envelope and envelope spectrum of the filtered signal. The rolling element damage inducing periodic impulses can be found in the zoomed version of the envelope. Additionally, the  $f_c$  (red dot) and the  $f_{BSF}$  and their harmonics (purple dots) are presented in the envelope spectrum. It can be concluded that the resonant frequency bands associated with the three different repetitive transients, including wheelset inducing impulses, bearing outer race defect inducing impulses, and rolling element defect inducing impulses, are automatically detected by the ACK method.

The results of the FK method are shown in Fig. 23(a), and the highest kurtosis value is located at level 3 with the center frequency 1562.5 Hz, the bandwidth 625 Hz. The envelope and envelope spectrum of this node are illustrated in the Fig. 23(b) and (c), respectively. The  $f_c$  (red dot),  $f_{BSF}$  and their harmonics (purple dots) can be observed in the envelope spectrum Fig. 23(c). It is convenient to believe the bearing has suffered the rolling element defect. However, it is hard to determine whether the bearing has an outer race defect.

The Fig. 24(a) shows the Autogram of the bearing multiple defects signal, and the largest kurtosis value appears at the node with the center frequency 468.75 Hz, the bandwidth 312.5 Hz. The envelope spectrum of the filtered signal is depicted in the Fig. 24(b). The envelope spectrum reveals a signature of wheel/wheel excitation with the rotation frequency  $f_r$  and its harmonics (indicated by red dots). On the contrary, bearing fault-related information cannot be observed from the envelope.



**Fig. 23.** (a) the Kurtogram of bearing multiple defects signal obtained by the FK; (b) the envelope of the filtered signal; (c) the envelope spectrum of the filtered signal.



**Fig. 24.** (a) the Autogram of the bearing multiple defects signal and (b) the envelope spectrum of the filtered signal.

## 6. Conclusion

This paper proposed a novel method to find different proper frequency bands of repetitive transients for achieving the wheelset-bearing systems' diagnosis. Firstly, the Fourier spectrum features are automatically explored by the SSR method and presented in the SSP. Secondly, the generation of the paving of the plane is adaptively achieved according to the SSP. Thirdly, the narrow-band signals are decomposed by the EWT. Fourthly, the correlated kurtosis is adopted to evaluate the impulsiveness of the filtered signals and used to generate the adaptive correlated Kurtograms. Finally, the frequency bands are selected from the adaptive correlated Kurtograms to extract the faulted-related repetitive transients, so as to identify the characteristic frequencies from the envelope spectrum.

The main advantage of the proposed method is its powerful ability to extract the interested periodic impulses from the complex vibration environment. This valuable achievement is effectively reached by calculating the SSP of the Fourier spectrum and correlated kurtosis of the filtered signals. The generation method of the paving of the plane can arbitrarily obtain the frequency boundary of each node, and can significantly improve the accuracy and rationality of the detected boundaries. Additionally, the correlated kurtosis indicator has a strong capability to highlight the specific periodic impulses from the multiple periodic impulses, thus it is very suitable for the wheelset-bearing systems' diagnosis.

The effectiveness of the proposed ACK method is demonstrated by the wheelset-bearing system's simulated signal and two measured signals. Meanwhile, comparisons with the FK and Autogram are conducted to illustrate the superiority of the proposed method.

## Declaration of Competing Interest

The author declare that there is no conflict of interest.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ymssp.2020.107511>.

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